This is the authors' version of the following article: Lee, H., Jin, S., & Lee, J. H. (2023). Ending the cycle of anxiety in language learning: A non-recursive path analysis approach. *System, 118*, 103154. The version of record is available at https://doi.org/10.1016/j.system.2023.103154. This version differs from the published article in layout and pagination; please cite the published version.

Ending the cycle of anxiety in language learning: A non-recursive path analysis approach

Hansol Lee[a], Seonghan Jin[a,*], Jang Ho Lee[b]

[a] *Department of English, Korea Military Academy, 574 Hwarang-ro, Nowon-gu, Seoul, 01805, Republic of Korea*

[b] *Department of English Education, Chung-Ang University, 84 Heukseok-ro, Dongjak-gu, Seoul, 06974, Republic of Korea*

[*] Corresponding author. *E-mail addresses:* hansol@kma.ac.kr (H. Lee), shjin0802@kma.ac.kr (S. Jin), jangholee@cau.ac.kr (J.H. Lee).

Abstract

From a socio-psychological perspective, learners with high second language (L2) learning anxiety are likely to avoid using L2, missing substantial L2 learning opportunities to improve their L2 competence, which would result in negatively impacting their L2 anxiety, forming a circulation process, also known as "vicious cycle of language anxiety" (Lou & Noels, 2020a). In this study, we explored the reciprocal relationship between self-perceived language competence, language anxiety, and language proficiency using two structural models applied to second language learning settings by recruiting 270 undergraduate students learning English as their foreign language, and building a recursive (i.e., a structural equation model) and a non-recursive (i.e., a path analysis with a feedback loop) structural models to investigate how to end the cycle of anxiety. Evidently, we found that there existed a reciprocal relationship between these factors, and building a growth language mindset, previous proficiency, and motivation would alleviate the "vicious cycle of language anxiety" in light of the results from the non-recursive path analysis model. Our findings are a major step forward from prior suggestions of piecemeal solutions that only partially affect a cyclical element to eventually break this so-called vicious cycle, such as fostering learners' growth mindset to mitigate their anxiety.

Keywords: Anxiety; Competence; Language mindset; Motivation; Non-recursive path analysis; Proficiency; Vicious cycle of language anxiety

1. Introduction

Through this study, we aimed to statistically validate what Lou and Noels' (2020a) termed the "vicious cycle of language anxiety," and provide evidence-based solutions to end this cycle by investigating the roles of predicting variables therein. In their introduction to the vicious cycle of language anxiety, Lou and Noels depict a common situation experienced by second language (L2) learners: Learners with high L2 anxiety are likelier to use the L2 to a lesser degree and to avoid L2 engagement, losing important opportunities to gain positive L2 use experience. Further, lack of engagement and experience leads to deprivation of opportunities to improve their L2 competence, further leading to an unpleasant language learning experience, which, in turn, intensifies their self-perception of a lack of language competence as well as their language learning anxiety. L2 learners with high L2 anxiety may find it challenging to break this cycle without receiving effective interventions. This "vicious cycle" is graphically illustrated in Fig. 1.

It follows that we must focus on how to help learners escape the cycle of anxiety. Lou and Noels (2020a) conducted an exemplary empirical study to tackle this issue. First, they confirmed that learners' perceived competence could mitigate their avoidance orientation, future contact avoidance, negative affect, and avoidance behavior; they further showed that learners' L2 anxiety, including their perceived rejection and avoidance orientation, negatively contributed to their L2 use. In other words, Lou and Noels identified how learners' perceived competence, anxiety, and use intertwine to create a vicious cycle of language anxiety. They then factored learners' language mindset (i.e., a belief that one's ability to learn L2 is immutable) into this mechanism to find that it predicted their L2 anxiety, particularly the perceived rejection aspect. Thus, they concluded that learners' growth language mindset (i.e., a belief that one's ability to learn the L2 is subject to change) contributed to learners' resilience in L2 learning (see Fig. 2).

In this study, we took Lou and Noels' mechanism further by incorporating a more intuitive and practical perspective. That is, we employed a non-recursive path analysis (i.e., a reciprocal relationship between variables in the model) with a feedback loop for the vicious cycle of L2 anxiety in order to statistically validate the structure of circulation between competence, anxiety, and L2 use. A layperson may not be able to catch a glimpse of any feedback loop structure in Lou and Noels' (2020a) statistical approach, wherein a regular structural equation model (SEM) was suggested (see Fig. 2). Instead, they checked a linear relationship from competence to L2 use through anxiety. Our approach (i.e., a non-recursive path analysis) should not be considered an attempt to denounce their model, but an alternative to such a common approach in the L2 field.

In doing so, we could suggest a comprehensive solution to end the vicious cycle of language anxiety by attempting to formulate a mechanism that can influence the entire feedback loop as opposed to focusing on its single components. We believe that validating Lou and Noels' mechanism helps us answer the question of *how* to escape the vicious cycle—that is, through fostering learners' growth language mindset, or one's belief that their language intelligence can be improved. Further, identifying and including additional exogenous (independent) variables for the remaining two elements of the vicious cycle of anxiety will constitute a more effective structural model for breaking the entire vicious cycle.

2. Literature review

In this section, we reviewed the roles of language anxiety and competence in L2 learning in order to examine how these variables lead learners into the vicious cycle of language anxiety. We also reviewed the literature dealing with solutions to this vicious cycle.

2.1. Roles of self-perceived competence and anxiety in defining the vicious cycle of anxiety

Learners' self-perception of their competence in L2 learning (e.g., Hashimoto, 2002; Khajavy, Ghonsooly, Hosseini Fatemi, & Choi, 2016; Yashima, 2002) and L2 anxiety (e.g., Hashimoto, 2002; Khajavy, Ghonsooly, Hosseini Fatemi, & Choi, 2016; MacIntyre & Charos, 1996) are among the two popularly studied factors whose close relationship is also well-established. Socio-psychological perspectives informed us that learners with high perceived L2 competence are likelier to believe that they can effectively handle negative and challenging situations in L2 communication, which will ultimately lower their negative emotions and prevent feelings of anxiety (Lee & Lee, 2020; Lou & Noels, 2016, Lou & Noels, 2017; Pyun, Kim, Cho, & Lee, 2014; Samimy & Tabuse, 1992). On the other hand, learners with low perceived L2 competence generally exhibit poor confidence in such discomforting situations, which increasing their anxiety in engaging in L2 communication. Jin and Lee's (2022) recent meta-analysis investigated such complex relationships between self-perceived competence, language anxiety, motivation, willingness to communicate (WTC), and frequency of language use. Based on 44 independent samples (N = 12,094), they found that learners' perceived competence had a significant contribution ($\beta = 0.43$) to the frequency of learners' target language use, along with other important factors such as language anxiety, motivation, and WTC (βs = −0.20, 0.21, 0.18, respectively). Further, learners' self-perceived competence was strongly correlated with learners' anxiety ($r = -0.41$) according to the meta-analytic SEM. Jin and Lee (2022) also identified a negative relationship of L2 anxiety with L2 WTC and L2 use. This negative association of anxiety with use and performance in L2 contexts was well-known

in earlier studies (e.g., Hashimoto, 2002; Horwitz, Horwitz, & Cope, 1986; Pyun, Kim, Cho, & Lee, 2014). This connection may be explained by contextual factors, such as learning environments and interlocutors, having a stronger effect on L2 learners' negative emotions or anxiety (MacIntyre & Charos, 1996; Peng & Woodrow, 2010; Yashima, 2002). That is, if L2 learners start to feel uncomfortable in L2 communication, they are likely to avoid unpleasant situations, such as being rejected or negatively judged by their instructors. These negative experiences, in turn, could make L2 learners avoid using the L2 frequently, which affects their ability and opportunities to practice L2 communication to improve their proficiency (Alrabai, 2022; Dewaele & MacIntyre, 2014; Horwitz, Horwitz, & Cope, 1986).

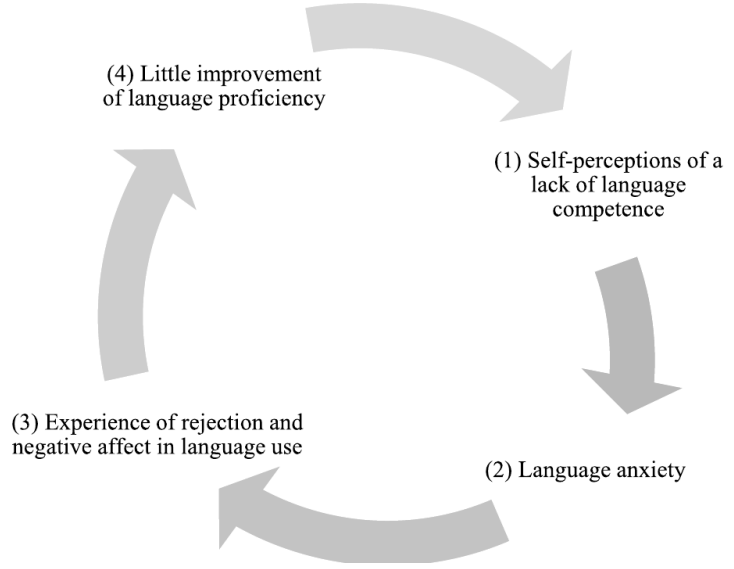


Fig. 1. Vicious cycle of language anxiety (adapted from Lou & Noels, 2020a, p. 2).

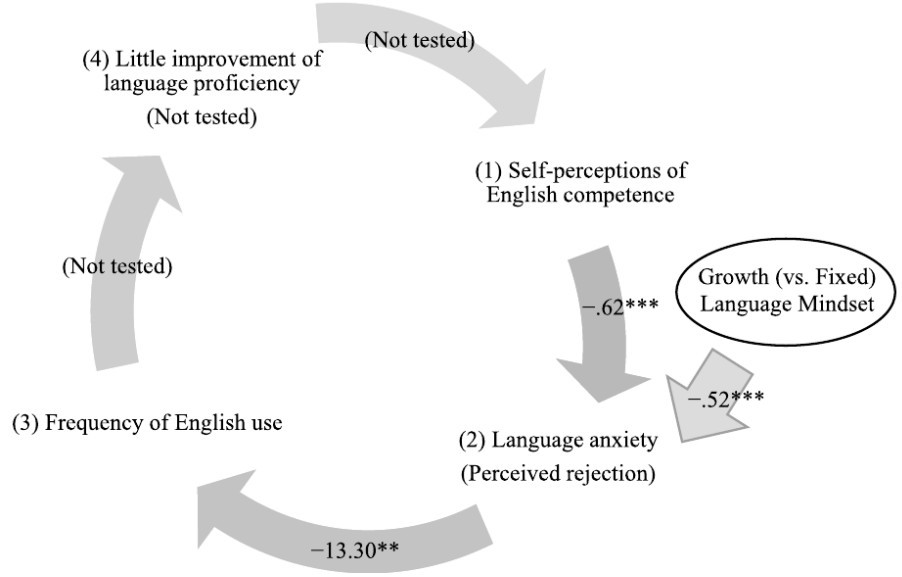


Fig. 2. Model of how learners' growth (vs. fixed) language mindset breaks the vicious cycle by mitigating their language anxiety (adapted from Lou & Noels, 2020a, p. 14).

Our choice of perceived competence and anxiety as variables of interest was inspired by Lou and Noels' (2020a) conceptualization of the vicious cycle of language anxiety, whereby language anxiety that arises from language learners' negative perception of their language competence, owing to negative use experiences (i.e., rejection and negative affect), leads to weak improvements in proficiency (i.e., language competence) as learners increasingly avoid engagement in the L2. As shown in Fig. 1 earlier, Lou and Noels empirically illustrated this sequence of events as a vicious loop that repeats again and again. They proposed that learners' language anxiety and perceived competence, which are negatively correlated with each other, could play an important and irreplaceable role in determining one's L2 use and proficiency.

L2 use and proficiency are the very core elements that allow "the use of language for meaningful communication in the process of foreign and second language acquisition" (Khajavy, Ghonsooly, Hosseini Fatemi, & Choi, 2016, p. 155) in communicative language teaching. Here, we turn to MacIntyre, Clément, Dörnyei, & Noels's (1998) model of variables affecting L2 WTC: L2 use is placed at the apex, implying that all the variables in the pyramid are antecedents of L2 WTC, and it ultimately leads to L2 use. The frequent use of L2, in turn, helps learners improve their proficiency. This model resonates not only with MacIntyre & Charos, 1996 statement that L2 learners must use the L2 in order to improve their proficiency (or "talk to learn" [p. 3]), but also with De Carli et al.'s (2015) empirical finding, derived from a cognitive approach, that language use affects proficiency.

We suppose that, if L2 learners do not communicate with others in the L2, they risk having limited opportunities to enhance their perceived competence and, thus, use of L2, which would lead to little improvement of L2 proficiency. Such learners will likely experience more anxiety about using the L2, creating greater discomfort in situations that require L2 communication that leads to a decline in their perceived L2 competence and, ultimately, lower L2 proficiency in a repetitive cycle.

2.2. Breaking the cycle of anxiety

Educators must find solutions to help L2 learners to break the vicious cycle of language anxiety in order to enhance learners' perception of competence and lower anxiety arising from negative L2 experiences. Thus, we pose the question, *How can we help L2 learners break this recursive loop?*

2.2.1. Affecting anxiety

Although scholars have identified self-confidence or competence (Ghasemi, 2018; Zarrinabadi, Rezazadeh, Karimi, & Lou, 2022) and motivation (Khodadady & Khajavy, 2013; Papi, 2010) as variables that could reduce L2 learners' anxiety, language mindset (i.e., a belief

that one's ability to learn L2 is immutable) has shown the strongest effect on learners' anxiety (Lou & Noels, 2020a, 2020b). Lou and Noels (2020a) suggested "growth mindset" (i.e., a belief that one's ability to learn L2 can be fostered and improved through learning and effort) as a solution to break the vicious cycle and improve competency and use of the L2 by assuming that a fixed language mindset led to learners' negative experiences and emotions (e.g., language-based rejection and future interaction avoidance). In a subsequent study, they argued that this growth mindset was especially useful for anxious L2 learners from immigrant backgrounds or novice learners pursuing intercultural communication (Lou & Noels, 2020b). Ozdemir and Papi (2022), in a similar vein, highlighted language mindsets as the origin of both anxiety and self-confidence in speaking the L2; they found that a fixed mindset was associated with anxiety about speaking the L2, while a growth mindset positively predicted L2 speaking self-confidence.

2.2.2. Affecting proficiency

As the cycle of language anxiety includes proficiency and competence, any solution must also consider factors that increase these elements. Affective variables such as motivation, attitude, perceived competence, anxiety, or grit, for example, are significant predictors of L2 communicative proficiency or performance (Lee & Hsieh, 2019). Since Gardner & Lambert, 1959 seminal study on motivational variables in second language acquisition, numerous scholars have identified L2 motivation as one of the most important variables for sustaining and developing L2 learning, and, consequently, attaining high L2 proficiency (e.g., Alamer & Alrabai, 2022; Dörnyei, 2005, 2009; Dornyei & Ryan, 2015; Gardner, 2010; Hashimoto, 2002). Khodadady & Khajavy, 2013 also confirmed that motivated L2 learners are likelier to have pleasant L2 learning experiences and sustain their L2 learning even in the face of challenging situations than those who are not motivated.

2.2.3. Affecting self-perceived competence

However, empirical efforts to identify instrumental variables (i.e., independent variables that only affect a dependent variable without affecting and being affected by other variables) to predict learners' perceived L2 competence are scarce, despite its importance in relation to other affective variables and its link to L2 use and proficiency (Clément, Baker, & MacIntyre, 2003; Hashimoto, 2002; Jin & Lee, 2022; MacIntyre & Charos, 1996). The vicious cycle of language anxiety (Lou & Noels, 2020a) shows that L2 learners improve their perceived L2 competence from positive, pleasant, and comfortable language experiences—even in difficult situations. Indeed, perceived L2 competence is associated with a good number of variables, such as learning strategies, motivation, anxiety, attitudes, and previous learning experiences; however, except for prior L2 proficiency, it is difficult to locate the most immediate

instrumental variable as other factors are correlated with each other and only indirectly contribute to self-perceived L2 competency (e.g., Green & Oxford, 1995; Moskovsky et al., 2016; Park, 1997; Taheri, Sadighi, Bagheri, & Bavali, 2019; Wu, 2010).

3. Model

3.1. Non-recursive structural model

Kline (2011) elucidates upon two types of models in SEM: (1) the commonly used recursive models wherein all causal effects are unidirectional (e.g., Lou & Noels' vicious cycle in Fig. 2) and (2) non-recursive models, which have feedback loops wherein more than two variables are specified as causes and effects of each other (e.g., when Figs. 1 and 3 are used as they are in structural models). The latter models have rarely been employed in the field of language learning, presumably because this type requires additional assumptions, such as of equilibrium and stationarity (Kline, 2011). To meet these assumptions, a feedback loop in non-recursive models should be a complete (or dampened out) causal process of the elements of the feedback loop, any reciprocal effects in the presumed feedback relation should be in a steady state after already being processed, and the reciprocal causal structure should not change over time (Heise, 1975; Kaplan et al., 2001; Kennyenny, 1979; as cited in Kline, 2011); however, there is no statistical procedure to evaluate these assumptions.

Another barrier to entry is to statistically construct a non-recursive structural model. Importantly, exogenous variables that should be "instrumental" variables to predict only one of the endogenous variables in a feedback loop are needed for model identification. Therefore, these instrumental variables should be evaluated before collecting data for the elements of a non-recursive structural model (Berry, 1984; Kline, 2011).

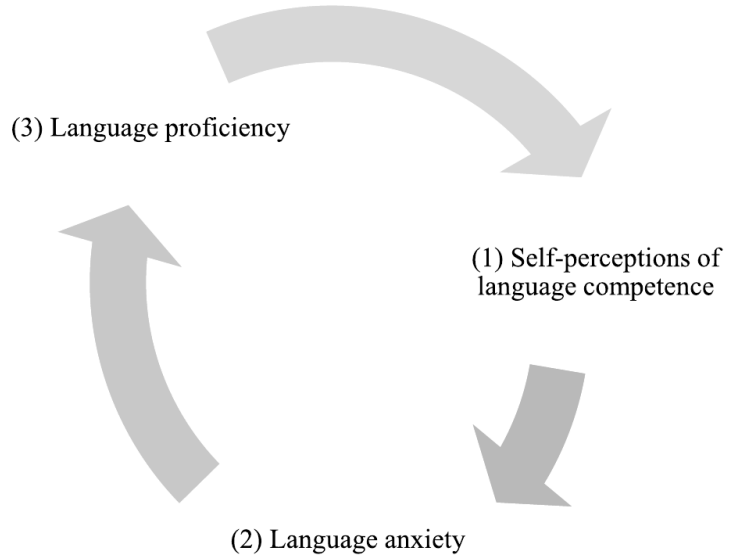


Fig. 3. A hypothesized vicious cycle of anxiety with language proficiency.

3.2. Application of the non-recursive structural model to the vicious cycle of anxiety

In this study, we employed a non-recursive path analysis with a feedback loop of learners' anxiety, self-perceived competence, and proficiency to confirm if our hypothesized circular-shaped model in Fig. 3 can be statistically validated. Unlike Lou and Noels' definition of the cycle, we replaced learners' L2 use with learners' L2 proficiency. We did so because, first, L2 anxiety, perceived L2 competence, and L2 proficiency have close mutual relations (MacIntyre, Noels, & Clément, 1997). Specifically, L2 anxiety is negatively associated with L2 proficiency (Teimouri et al., 2019), whereas perceived L2 competence is positively associated with L2 proficiency (Ghasemi, 2018; Khajavy, Ghonsooly, Hosseini Fatemi, & Choi, 2016). It is more intuitive and practical to evaluate learners' L2 proficiency than their frequency of L2 use when examining any recognizable improvement of L2 learning from assessment perspectives; that is, L2 proficiency can be more objective and stable than the frequency of L2 use (which is often measured via the participants' self-reported questionnaire items) in terms of measurement. Second, it is meaningful to include L2 proficiency as a dependent variable with relation to growth mindset, as it is influenced by the frequent use of L2 and the ultimate target variable of L2 learning. This approach should help us investigate a more long-term effect of growth mindset than what Lou and Noels (2020a) presented.

Unlike the normative approach to path analysis—namely, recursive models—wherein all effects are unidirectional, a non-recursive path analysis model has a feedback loop that includes bidirectional relationships (Kline, 2011). To statistically identify non-recursive models, variables in the feedback loop need their own instrumental variables that only affect their target variables without affecting other variables. In view of Kline (2011), we include three variables that affect each of the three elements of the vicious cycle of language anxiety

and these variables do not directly affect the other elements. Thus, we suggest (1) growth language mindset, (2) motivation, and (3) prior English proficiency as instrumental variables for (1) anxiety, (2) current English proficiency, and (3) perceived competence, respectively. Fig. 4 represents our hypothesized non-recursive path analysis model.

We thus summarize our objective in two research questions:

> Research question 1. Is Lou and Noels' (2020a) vicious cycle of language anxiety valid among South Korean learners of English as foreign language? If so, does the learners' growth language mindset break the vicious cycle by mitigating their anxiety?
>
> Research question 2. Do the collected data successfully fit the hypothesized non-recursive model for the vicious cycle of anxiety? If so, how is the non-recursive model being operated to end the vicious cycle of anxiety?

4. Methods

4.1. Participants

We initially recruited 298 undergraduate students learning English as foreign language (EFL) registered at the mandatory English course in several universities located in Seoul, Republic of Korea. A survey revealed that their average age was 19.2 years (i.e., 18 to 20) at the time of the study, and a 7:3 male–female ratio. On average, the length of their English study abroad experience was less than 2 months (M = 1.93). The learners who participated in our study can be considered EFL learners given their limited exposure to English by the time of the study. For the data analyses, 270 students were finally included after 28 students opted to withdraw from the study during the semester. Regarding the average proficiency level of the participants, they were "Independent Users" (between CEFR B1 and B2 levels) according to the CEFR benchmarks (the detailed information about their proficiency test scores are described in the results section).

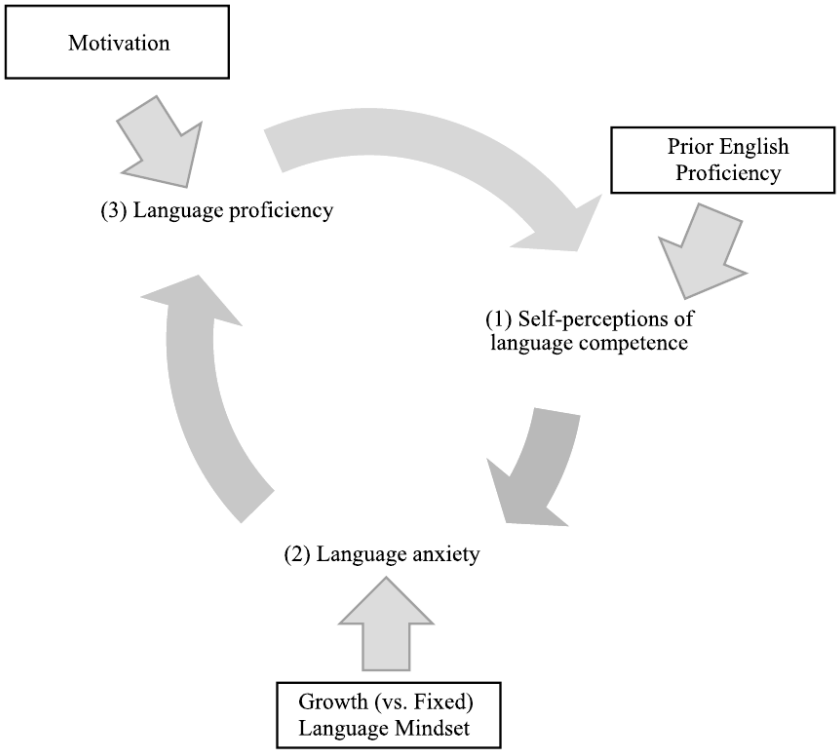


Fig. 4. A hypothesized non-recursive path analysis model for the vicious cycle of anxiety.

4.2. Procedure

An earlier version of the survey was piloted with 32 undergraduate EFL students to check if questionnaire items across the five sections were comprehensible in the first language of the respondents. According to the pilot study, the survey took about 10 min; an earlier version was revised based on the responses of the respondents. Data collection began after any subsequent revisions to the survey.

At the beginning of an academic semester, we recruited South Korean EFL students at several universities in Seoul. Two hundred and ninety-eight students responded to the survey across five sections. In the first section, survey respondents were instructed to read the overall procedure of the survey, sign the consent form, and leave their personal contact information (without any indication of their identification) to receive monetary compensation for their participation. In the second section, respondents were asked to report their demographic information, such as age, gender, and length of study abroad in English-speaking countries. In the third and the fourth (the last) sections, respondents were asked to answer a series of questionnaire items to measure their language mindset status and motivation, respectively. After completing the survey, they were required to take English Testing Service's (ETS, 2019) TOEIC® Listening and Reading Test (visit here for an official sample of the test) to measure their (prior) English proficiency.

At the end of the academic semester, we contacted the survey respondents and asked them to respond to an additional survey with two sections for their perceived L2 competence and L2 anxiety. They were required to take ETS's TOEIC® Speaking Test (visit here for an official sample of the test) to measure their (current) English proficiency thereafter. After the final phase of data collection, 28 participants had not completed the entire procedure. In the end, 270 participants' data were empirically analyzed to derive the results of the study.

4.3. Instruments

The two surveys consisted of four and two sections, respectively, and were designed to measure four important constructs: (1) items from the Language Mindset Inventory, (2) L2 motivation, (3) self-perceived L2 competence, and (4) L2 anxiety. We employed TOEIC® Listening and Reading Test and ETS's TOEIC® Speaking Test to measure learners' prior and current English proficiency, respectively.

4.3.1. Growth (vs. Fixed) language mindset

To measure learners' language mindset in L2 learning contexts, we adapted Lou & Noels, 2017 Language Mindset Inventory, which includes 18 questions across 3 constructs—namely, general language intelligence beliefs, L2 aptitude beliefs, and age-sensitivity beliefs about language learning—on a six-point Likert scale. Following Jin and Lee (2022) and Lou and Noels (2020a), we coded learners' responses (fixed aspects were reverse coded) and combined them to compute growth (vs. fixed) language mindset indices. This way, positive and negative values represented growth and fixed language mindset status, respectively. The sample reliability (Cronbach's alpha) of the three composite items—general language intelligence beliefs, L2 aptitude beliefs, and age-sensitivity beliefs about language learning items—was 0.80.

4.3.2. Second language learning motivation

To measure learners' L2 motivation, particularly in the context of assessing their language mindset status, we adopted Noels, Clément, & Pelletier, 1999 nine survey items for the intrinsic motivation aspect. Their motivation scale was developed in view of the self-determination theory of Deci & Ryan, 1985, who claimed that intrinsic and extrinsic motivational aspects best represent learners' motivational orientation of self-determined performance. Between these two types of motivation concepts, we focused on the former in view of the Noels, Clément, & Pelletier, 1999 empirical finding that intrinsic motivation was more closely related to language learning outcome. The sample reliability (Cronbach's alpha) of the nine intrinsic motivation questionnaire items was 0.92.

4.3.3. Self-perceived second language competence

To measure learners' self-perceptions of English competence, the participants were asked to self-evaluate their English competence in speaking, listening, reading, and writing dimensions on a seven-point Likert scale, similar to the third study conducted in Lou and Noels' (2020a) work. The sample reliability (Cronbach's alpha) of this measure was 0.89.

4.3.4. Second language learning anxiety

To measure learners' L2 anxiety, we adopted a shorter version of Horwitz, Horwitz, & Cope, 1986 Foreign Language Classroom Anxiety Scale, also known as the S-FLCAS. It consists of eight items measured on a five-point Likert scale, and was first suggested by MacIntyre (1992) and later stipulated by Dewaele & MacIntyre, 2014. The sample reliability (Cronbach's alpha) of this measure was 0.89.

4.3.5. Prior and current English proficiency

To measure learners' prior English proficiency at the beginning of the semester, learners were required to take the ETS's TOEIC® Listening and Reading Test. As this construct was measured to predict learners' self-perceived L2 proficiency in general, we concluded that an English proficiency test that is designed to assess an overall level of one's English performance could be an appropriate instrument.

To measure learners' current English proficiency at the end of the semester, we employed the ETS's TOEIC® Speaking Test, as we surmised that learners' L2 anxiety would be more related to the production aspect of L2 use. Therefore, assessing their speaking 6 capabilities was considered more appropriate in the context of the vicious cycle of language anxiety.

4.4. Data analysis

We used STATA Version 16 (StataCorp, 2019) to replicate Lou and Noels' (2020a) SEM model (Fig. 2) via SEM and to build a non-recursive path analysis model by fitting the hypothesized model (Fig. 4) with the collected data to answer our research questions. Unlike a typical SEM model, also known as a recursive model, non-recursive models have not been employed sufficiently in the field of L2 learning. This meant that we carefully followed the approaches established in the literature (Fan et al., 2016; Kline, 2011), including Toker and Akbay's (2022) instructions, to successfully measure complex relationships among the constructs. We particularly focused on the reciprocal effects in the vicious cycle.

Before fitting the SEM model as well as the path analysis model, we checked the descriptive statistics of the target constructs, such as mean, standard deviation, skewness, kurtosis, and bivariate correlation. For the SEM model (Fig. 2), we checked if each latent variable had been

successfully measured by its indicators to confirm the construct validity of latent variables.

Regarding the non-recursive path analysis model (see Fig. 4), instead of constructing latent variables, we computed the averages for the observed variables for the full path analysis model in order to ensure enough statistical power according to the sample size of the present study. Thereafter, we computed the path analysis model to answer the research questions.

When evaluating goodness of fit for both SEM and path analysis models, we adopted the following four indices, per the literature (e.g., Jin & Lee, 2022; Kline, 2011; Lee, Warschauer, & Lee, 2020, Lee, Jung, & Lee, 2022): (1) chi-square test ($p > 0.05$ or $\chi^2/df < 5$ for acceptable fit); (2) root mean square error of approximation (RMSEA <0.08 for acceptable fit); (3) comparative fit index (CFI >0.90 for acceptable fit); and (4) Tucker–Lewis index (TLI >0.90 for acceptable fit).

5. Results

5.1. Descriptive statistics

Table 1 presents the descriptive statistics, such as means, standard deviations, 95% CI, internal reliabilities (when applicable), and normality test results, for the six observed variables for the path analysis model. The six observed variables passed the Jarque and Bera normality test (1987) either at the 1% or 5% significance level. Of the six variables, four, namely, growth (vs. fixed) language mindset, motivation, anxiety, and self-perceived competence, were computed by averaging multiple items. All had acceptable levels of internal reliability ($\alpha >$ 0.80).

With respect to the growth (vs. fixed) language mindset, about 14 students had overall fixed language mindset (<0), 13 students had overall mixed language mindset (=0), and the rest ($n =$ 243) had overall growth language mindset (>0). Regarding motivation, anxiety, and self-perceived competence, learners' responses ranged from minimum to maximum scores for each measurement, and the mean of each instrument indicated that the learners' average was above the middle scores, at 3.47 (>2.5), 2.88 (>2.5), and 5.22 (>4.0) for motivation, anxiety, and self-perceived competence, respectively. Regarding their prior and current English proficiency test scores (i.e., the average results of their TOEIC® Listening and Reading Test [823.18] and Speaking Test [139.07]), participants could be considered, on average, "Independent Users" (between CEFR B1 and B2 level scales; ETS, 2019) of English.

5.2. Research question 1. Breaking the vicious cycle of language anxiety with growth language mindset

To replicate Fig. 2, as suggested by Lou and Noels (2020a), we utilized learners' growth (vs. fixed) language mindset, self-perceived competence, anxiety, and current English proficiency to build a SEM model. We also included an interaction term between the growth language mindset and self-perceived competence. As we employed a shorter version of Horwitz, Horwitz, & Cope, 1986 anxiety scale, we could not separate our anxiety factor into two constructs for "perceived rejection" and "avoidance orientation." Nevertheless, the results indicated that our collected data successfully replicated Lou and Noels' SEM model, as shown in Fig. 5.

With respect to the path coefficients of our interest, data analysis confirmed the vicious cycle. That is, learners' perceived competence had a negative effect on their anxiety ($\beta = -0.49$, $SE = 0.05$, $p < 0.001$), which, in turn, negatively influenced their English proficiency ($\beta = -0.47$, $SE = 0.06$, $p < 0.001$). Thus, when learners had low perceived competence in L2, they tended to have higher L2 anxiety and lower English proficiency, whereas their peers with high perceived competence had lower anxiety and higher English proficiency.

Table 1
Descriptive statistics of the observed variables for the path analysis model.

Variable	Mean	SD	Min	Max	α (# of items)	Normality test[a]
Growth (vs. fixed) language mindset	2.25	1.51	−2.67	5.00	0.80 (18)	Passed at 1%
Motivation	3.47	0.83	1.00	5.00	0.92 (9)	Passed at 5%
Prior English proficiency (L/R)	823.18	81.84	605	990	N/A	Passed at 5%
Anxiety	2.88	0.84	1.00	4.75	0.89 (.8)	Passed at 5%
Current English proficiency (SPK)	139.07	17.75	80	200	N/A	Passed at 1%
Self-perceived competence	5.22	1.51	1.00	7.00	0.89 (4)	Passed at 5%

[a] Jarque & Bera, 1987 test for normality.

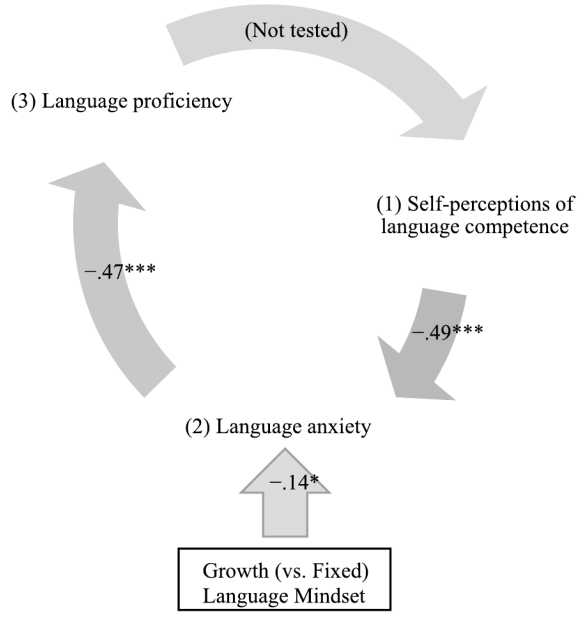


Fig. 5. Results of structural equation modeling replicating Lou and Noels' model to confirm if learners' growth (vs. fixed) language mindset breaks the vicious cycle by mitigating their language anxiety.
Note. The goodness of model fit was confirmed, as follows: (1) chi-square test had a *p*-value of zero but χ^2/df ratio was 2.36 (<5 for acceptable fit); (2) RMSEA was 0.07 (<0.08 for acceptable fit); (3) CFI was 0.94 (>0.90 for acceptable fit); and (4) TTLI was 0.92 (>0.90 for acceptable fit).

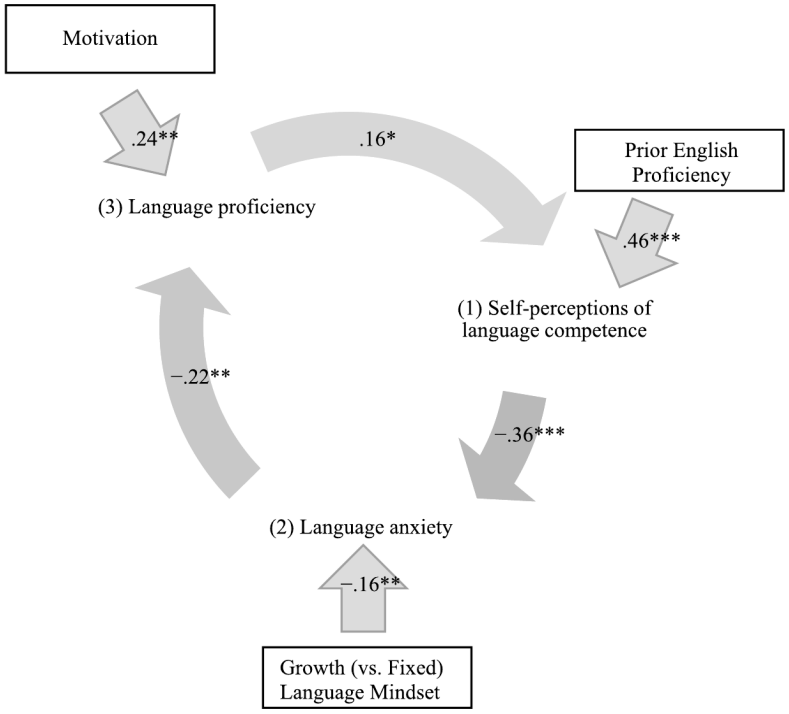


Fig. 6. Results of non-recursive path analysis model for the vicious cycle of anxiety.
Note. The goodness of model fit was confirmed, as follows: (1) chi-square test had a *p*-value of 0.07 (<0.05 for acceptable fit) and χ^2/df ratio was 2.34 (<5 for acceptable fit); (2) RMSEA was 0.07 (<0.08 for acceptable fit); (3) CFI was 0.99 (>0.90 for acceptable fit); and (4) TTLI was 0.94 (>0.90 for acceptable fit).

Although the interaction between growth (vs. fixed) language mindset and perceived competence did not have a statistically significant effect on learners' anxiety (β = −0.04, *SE* = 0.06, *p* = 0.55), their growth (vs. fixed) language mindset mitigated anxiety (β = −0.14, *SE* = 0.05, *p* = 0.01). Thus, if learners could raise their awareness of growth language mindset, this would mitigate their L2 anxiety and contribute to enhancing their English proficiency. We thus showed that learners' growth language mindset can play an important role in breaking the vicious cycle of language anxiety.

5.3. Research question 2. Ending the vicious cycle of language anxiety in a feedback loop

In our second research question, we hypothesized that influencing each element of the vicious cycle of language anxiety would be more effective in ending (than breaking) the cycle. To this end, we included learners' motivation and prior English proficiency as instrumental variables for the remaining two elements of the vicious cycle, namely, current English proficiency and self-perceived competence. Fig. 6 illustrates the results of our non-recursive path analysis model.

Regarding the path coefficients in the path analysis model, we first confirmed that our data successfully fit the model of the vicious cycle of language anxiety. As shown in the previous SEM model (Fig. 3), we found that learners' perceived competence had a negative effect on their anxiety (β = −0.36, *SE* = 0.05, *p* < 0.001), which then negatively influenced their English proficiency (β = −0.22, *SE* = 0.08, *p* = 0.01). As this path analysis model included a feedback loop, learners' English proficiency influenced their perceived competence (β = 0.16, *SE* = 0.08, *p* = 0.04) in the feedback loop as well. Notably, the previous SEM model only showed a linear relationship, from competence to proficiency through anxiety, whereas our path analysis model illustrates a cyclical relationship of the three elements—a more intuitive approach.

To verify if learners' growth (vs. fixed) language mindset played an important role in breaking the vicious cycle of language anxiety, we checked its path coefficient to its predicting variable (i.e., anxiety). There was a statistically significant coefficient of −0.16 (*SE* = 0.06, *p* = 0.004). To investigate if we could actually end the vicious cycle by influencing the remaining elements of the cycle, namely, English proficiency and self-perceived competence, we checked the path coefficients of their instrumental variables.

Evidently, learners' motivation had a statistically significant effect on their current English proficiency (β = 0.24, *SE* = 0.08, *p* = 0.002), and their prior English proficiency significantly affected their perceived competence (β = 0.46, *SE* = 0.06, *p* < 0.001). Taken together, we confirmed that the data and subsequent analyses supported our hypotheses; that is, a vicious cycle of language anxiety existed among the participants; however, influencing both learners' growth (vs. fixed) language mindset and motivation could break this cycle. We thus showed

that not only does the growth language mindset *break* the vicious cycle by mitigating learners' anxiety, but that this effect is further enhanced by influencing both the growth language mindset and motivation.

6. Discussion and conclusion

We validated the Lou and Noels' (2020a) vicious cycle of language anxiety using a sample of South Korean undergraduate EFL learners and by replacing the frequency of learners' L2 use with their L2 proficiency. To extend Lou and Noels' recursive SEM model to our context, we built a structural model based in non-recursive path analysis. This model was equipped with a feedback loop for the vicious cycle, which was identified by each element's instrumental variables. In this section, the main findings are further discussed, such that we validated the existence of the vicious cycle of language anxiety equipped with our sample based on the non-recursive model with a feedback loop and confirmed the remedial role of growth language mindsets in mitigating learners' anxiety as well as that of motivation in improving learners' proficiency to help learners escape the vicious cycle of anxiety.

6.1. Validating Lou and Noels' vicious cycle of language anxiety

We confirmed the existence of the vicious cycle of language anxiety among the sampled undergraduate EFL learners. In agreement with established findings, learners' perceived L2 competence was significantly related to their L2 anxiety (Jin & Lee, 2022), which, in turn, negatively predicted their L2 proficiency (Teimouri et al., 2019). We further confirmed that the construct of learners' growth language mindset mitigated the negative effect of anxiety, thus helping learners improve their L2 proficiency as well as their perceived competence in L2 learning.

By extending Lou and Noels' model to our sample, we confirmed its validity to different populations. Our results, which were in line with Lou and Noels' findings, further confirmed the importance of the growth language mindset as a remedy to break this vicious cycle. As a result, we enriched the literature by including learners' proficiency (as an achievement-related learner factor) as the dependent variable over frequency of L2 use when exploring the long-term effect of growth language mindset in the vicious cycle. Through our model, we extended Lou and Noels' proposition regarding the role of growth language mindset in L2 learning.

6.2. Ending the vicious cycle of anxiety

In addition to investigating a linear pattern of the vicious cycle, we also statistically examined the cyclic feature of the vicious cycle for a more intuitive understanding of the subject. The results of the non-recursive path analysis model showed that focusing on learners' growth

language mindset, motivation, and English proficiency simultaneously would be more effective in breaking the vicious cycle. We surmise that this synergic effort can end the vicious cycle of anxiety in comparison to piecemeal solutions that only partially concern one cyclical element.

Overall, our model helped us understand the mechanism underlying the vicious cycle of anxiety better and offered an evidence-based perspective on the means to breaking the vicious cycle. For example, Kim & Kim, 2014 argued that one's motivation significantly influenced their proficiency; similarly, we highlighted the exclusive role of motivation as an instrumental variable for learners' proficiency within the framework of the vicious cycle of language anxiety.

As another component in our model, learners' growth language mindset appears to influence their proficiency, albeit indirectly; given its close association with motivation (Lou & Noels, 2019a), the role of growth mindset should receive more attention in the literature. Taken together, educators are strongly encouraged to find pedagogical strategies to improve both learners' growth language mindset and motivation when aiming to address the vicious cycle of language anxiety. For example, when it comes to empirical evidence on how to promote growth language mindset, Lou & Noels, 2016 found that reading an article which promotes growth language mindsets led learners to be more resilience in their L2 learning. Similarly, Lee, Lee, & Scarcella, 2023 confirmed that a 10-week English course where learners read chapters of a growth mindset related book had an overall positive impact of learners' L2 achievement. With respect to how to improve one's motivation, due to the space limitation readers are encouraged to see related meta-analytic efforts (e.g., Al-Hoorie, 2018; Masgoret & Gardner, 2003) as there are ample empirical studies on this topic unlike language mindset.

6.3. Implications

In this study, our findings indicate the importance of focusing on learners' growth language mindset and motivation in terms of pedagogical implications. Among the three instrumental variables we examined, growth language mindset and motivation should be prioritized when addressing the vicious cycle because prior English proficiency cannot be influenced or manipulated in the classroom. In doing so, instructors can design courses to enhance learners' growth mindset (e.g., Lee, Lee, & Scarcella, 2023; Lou & Noels, 2016) and motivation (e.g., Sato, 2020; Sato & Lara, 2019; see Al-Hoorie, 2018; Masgoret & Gardner, 2003 for meta-analytic reviews). As suggested by Lou and Noels (2019b), promoting learners' growth mindset could be achieved by employing explicit interventions and making suitable classroom interactions with learners. Regarding the former, L2 instructors can include articles on growth mindset (e.g., Dweck, 2006) in their classes, read them together with learners, and discuss

their value in the L2 learning process, which has been shown to be an effective way to promote growth mindset (e.g., Lee, Lee, & Scarcella, 2023; Lou & Noels, 2016). L2 instructors can also engage in growth mindset-promoting classroom interactions "by helping students set mastery goals and making attributions focused on the learning process, helping them feel safe to risk making mistakes, and providing them with constructive feedback to improve" (Lou & Noels, 2019b, p. 9). Given the propositions of the implicit theory of mindset (Leith et al., 2014), we believe that such efforts (particularly helping learners to set mastery goals) are also likely to promote learning motivation and thus help break the vicious cycle of language anxiety.

With respect to the research methods implication, we employed a relatively unfamiliar statistical approach, namely, a non-recursive structural model, for a more intuitive solution to breaking the vicious cycle of language anxiety. We hope that our description and use of this approach will encourage other L2 researchers to consider adopting such non-recursive models for examining various cyclic relationships (i.e., feedback loops) among important variables in L2 learning. To this end, we spared a separate section to carefully introduce the definition, theoretical assumptions, and statistical rules related to non-recursive structural models. We are confident that our attempt contributes to expanding methodological avenues, particularly for future research on individual learner differences (e.g., motivation, anxiety, L2 WTC, L2 learning strategies, L2 aptitude, and age).

6.4. Limitations

Our study is not without limitations. First, a generalization of our findings to wider populations needs further validation through studies based in diverse contexts. Our sample only included South Korean undergraduate EFL learners, while Lou and Noels' (2020a) study sampled undergraduate students of English as a second language in Canada. We recommend expanding to sample to different ages (e.g., children, adolescents, etc.) and language groups (e.g., learners of L2 other than English).

Second, meeting assumptions and statistical requirements for non-recursive structural models was challenging and may invite criticism of our model-building. For example, our selection of instrumental variables may be challenged—in terms of practice or theory. Our statistical compromise to build a path analysis model instead of an SEM model in order to secure enough statistical power may have compromised the accuracy in estimating our constructs. Therefore, we recommend a re-evaluation of our methodology using a larger sample size and more sophisticated research design to validate our findings and extend our understanding of the vicious cycle of language anxiety and how to end it.

Funding

This work was supported by the Hwarangdae Research Institute at the Korea Military Academy (22-군학-9).

Author statement

Hansol Lee: Conceptualization, Methodology, Software, Funding acquisition, Writing-Original Draft.

Seonghan Jin: Writing-Original Draft, Data Curation.

Jang Ho Lee: Writing-Review and Editing, Validation.

Declaration of competing interest

None.

Acknowledgement

An earlier version of the manuscript was presented at the 2022 Applied Linguistics Association of Korea (ALAK) International Conference.

References

Alamer, A., & Alrabai, F. (2022). The causal relationship between learner motivation and language achievement: New dynamic perspective. *Applied Linguistics, 44*(1), 148–168. https://doi.org/10.1093/applin/amac035

Al-Hoorie, A. H. (2018). The L2 motivational self system: A meta-analysis. *Studies in Second Language Learning and Teaching, 8*(4), 721–754. https://doi.org/10.14746/ssllt.2018.8.4.2

Alrabai, F. (2022). The predictive role of anxiety and motivation in L2 proficiency: An empirical causal model. *Language Teaching Research. Advance Online Publication*. https://doi.org/10.1177/13621688221136247

Berry, W. D. (1984). *Nonrecursive causal models*. Sage Publications.

Clément, R., Baker, S. C., & MacIntyre, P. D. (2003). Willingness to communicate in a second language: The effects of context, norms, and vitality. *Journal of Language and Social Psychology, 22*(2), 190–209. https://doi.org/10.1177/0261927X03022002003

De Carli, F., Dessi, B., Mariani, M., Girtler, N., Greco, A., Rodriguez, G., … Morelli, M. (2015). Language use affects proficiency in Italian–Spanish bilinguals irrespective of age of second language acquisition. *Bilingualism: Language and Cognition*, *18*(2), 324–339. https://doi.org/10.1017/S1366728914000054

Deci, E. L., & Ryan, R. M. (1985). The general causality orientations scale: Self-determination in personality. *Journal of Research in Personality, 19*(2), 109–134. https://doi.org/10.1016/0092-6566(85)90023-6

Dewaele, J. M., & MacIntyre, P. D. (2014). The two faces of Janus? Anxiety and enjoyment in the foreign language classroom. *Studies in Second Language Learning and Teaching, 4*(2), 237–274.

Dörnyei, Z. (2005). *The psychology of the language learner: Individual differences in second language acquisition*. Lawrence Erlbaum Associates Publishers. https://doi.org/10.4324/9781410613349

Dörnyei, Z. (2009). The L2 motivational self-system. In Z. Dörnyei, & E. Ushioda (Eds.), *Motivation, language identity and the L2 self* (pp. 9–42). Multilingual Matters.

Dornyei, Z., & Ryan, S. (2015). *The psychology of the language learner revisited*. Routledge. https://doi.org/10.4324/9781315779553

Dweck, C. S. (2006). *Mindset: The new psychology of success*. Random House.

English Testing Service. (2019). *Mapping the TOEIC® tests on the common European framework of reference for languages*. https://www.ets.org/content/dam/ets-org/pdfs/toeic/toeic-mapping-cefr-reference.pdf.

Fan, Y., Chen, J., Shirkey, G., John, R., Wu, S. R., Park, H., & Shao, C. (2016). Applications of structural equation modeling (SEM) in ecological studies: An updated review. *Ecological Processes, 5*(1), 1–12. https://doi.org/10.1186/s13717-016-0063-3

Gardner, R. C. (2010). *Motivation and second language acquisition: The socio-educational model*. Peter Lang Publishing.

Gardner, R. C., & Lambert, W. E. (1959). Motivational variables in second-language acquisition. *Canadian Journal of Psychology, 13*(4), 266–272.

Ghasemi, A. A. (2018). Ideal L2 self, visual learning styles, and L2 self confidence in predicting language proficiency and L2WTC: A structural equation modeling. *English Teaching & Learning, 42*(2), 185–205. https://doi.org/10.1007/s42321-018-0010-8

Green, J. M., & Oxford, R. (1995). A closer look at learning strategies, L2 proficiency, and gender. *TESOL Quarterly, 29*(2), 261–297. https://doi.org/10.2307/3587625

Hashimoto, Y. (2002). Motivation and willingness to communicate as predictors of reported L2 use: The Japanese ESL context. *Second Language Studies, 20*(2), 29–70. Retreived from https://core.ac.uk/download/pdf/77238697.pdf.

Horwitz, E. K., Horwitz, M. B., & Cope, J. (1986). Foreign language classroom anxiety. *The Modern Language Journal, 70*(2), 125–132. https://doi.org/10.2307/327317

Jarque, C. M., & Bera, A. K. (1987). A test for normality of observations and regression residuals. *International Statistical Review, 55*(2), 163–172. https://doi.org/10.2307/1403192

Jin, S., & Lee, H. (2022). Willingness to communicate and its high-evidence factors: A meta-analytic structural equation modeling approach. *Journal of Language and Social Psychology, 41*(6), 716–745. https://doi.org/10.1177/0261927X221092098

Khajavy, G. H., Ghonsooly, B., Hosseini Fatemi, A., & Choi, C. W. (2016). Willingness to communicate in English: A microsystem model in the Iranian EFL classroom context. *TESOL Quarterly, 50*(1), 154–180. https://doi.org/10.1002/tesq.204

Khodadady, E., & Khajavy, G. H. (2013). Exploring the role of anxiety and motivation in foreign language achievement: A structural equation modeling approach. *Porta Linguarum, 20*, 269–286.

Kim, T. Y., & Kim, Y. K. (2014). A structural model for perceptual learning styles, the ideal L2 self, motivated behavior, and English proficiency. *System, 46*, 14–27. https://doi.org/10.1016/j.system.2014.07.007

Kline, R. B. (2011). *Principles and practice of structural equation modeling* (3rd ed.). Guilford Press.

Lee, H., Jung, G., & Lee, J. H. (2022). Simple view of second language reading: A meta-analytic structural equation modeling approach. *Scientific Studies of Reading, 26*(6), 585–603. https://doi.org/10.1080/10888438.2022.2087526

Lee, H., Lee, J. H., & Scarcella, R. C. (2023). Influencing language mindsets and English writing competence through an EAP program: A longitudinal study with latent transition analysis. *Language Teaching Research. Online Advance Publication*. https://doi.org/10.1177/13621688221143258

Lee, H., Warschauer, M., & Lee, J. H. (2020). Toward the establishment of a data-driven learning model: Role of learner factors in corpus-based second language vocabulary learning. *The Modern Language Journal, 104*(2), 345–362. https://doi.org/10.1111/modl.12634

Lee, J. S., & Hsieh, J. C. (2019). Affective variables and willingness to communicate of EFL learners in in-class, out-of-class, and digital contexts. *System, 82*, 63–73. https://doi.org/10.1016/j.system.2019.03.002

Lee, J. S., & Lee, K. (2020). Affective factors, virtual intercultural experiences, and L2 willingness to communicate in in-class, out-of-class, and digital settings. *Language Teaching Research, 24*(6), 813–833. https://doi.org/10.1177/1362168819831408

Leith, S. A., Ward, C. L. P., Giacomin, M., Landau, E. S., Ehrlinger, J., & Wilson, A. E. (2014). Changing theories of change: Strategic shifting in implicit theory endorsement. *Journal of Personality and Social Psychology, 107*(4), 597–620. https://doi.org/10.1037/a0037699

Lou, N. M., & Noels, K. A. (2016). Changing language mindsets: Implications for goal orientations and responses to failure. *Contemporary Educational Psychology, 46*, 22–33. https://doi.org/10.1016/j.cedpsych.2016.03.004

Lou, N. M., & Noels, K. A. (2017). Measuring Language mindsets and modeling their relations with goal orientations and emotional and behavioral responses in failure situations. *The Modern Language Journal, 101*(1), 214–243. https://doi.org/10.1111/modl.12380

Lou, N. M., & Noels, K. A. (2019a). Language mindsets, meaning-making, and motivation. In M. Lamb, K. Csizér, A. Henry, & S. Ryan (Eds.), *The palgrave handbook of motivation for language learning* (pp. 537–559). Palgrave Macmillan. https://doi.org/10.1007/978-3-030-28380-3_26.

Lou, N. M., & Noels, K. A. (2019b). Promoting growth in foreign and second language education: A research agenda for mindsets in language learning and teaching. *System, 86*, Article 102126. https://doi.org/10.1016/j.system.2019.102126

Lou, N. M., & Noels, K. A. (2020a). Breaking the vicious cycle of language anxiety: Growth language mindsets improve lower-competence ESL students' intercultural interactions. *Contemporary Educational Psychology, 61*, Article 101847. https://doi.org/10.1016/j.cedpsych.2020.101847

Lou, N. M., & Noels, K. A. (2020b). Mindsets matter for linguistic minority students: Growth mindsets foster greater perceived proficiency, especially for newcomers. *The Modern Language Journal, 104*(4), 739–756. https://doi.org/10.1111/modl.12669

MacIntyre, P. D. (1992). *Anxiety, language learning and stages of processing [Unpublished doctoral dissertation]*. University of Western Ontario.

MacIntyre, P. D., & Charos, C. (1996). Personality, attitudes, and affect as predictors of second language communication. *Journal of Language and Social Psychology, 15*(1), 3–26. https://doi.org/10.1177/0261927X960151001

MacIntyre, P. D., Clément, R., Dörnyei, Z., & Noels, K. A. (1998). Conceptualizing willingness to communicate in a L2: A situational model of L2 confidence and affiliation. *The Modern Language Journal, 82*(4), 545–562. https://doi.org/10.1111/j.1540-4781.1998.tb05543.x

MacIntyre, P. D., Noels, K. A., & Clément, R. (1997). Biases in self-ratings of second language proficiency: The role of language anxiety. *Language Learning, 47*(2), 265–287. https://doi.org/10.1111/0023-8333.81997008

Masgoret, A. M., & Gardner, R. C. (2003). Attitudes, motivation, and second language learning: A meta-analysis of studies conducted by gardner and associates. *Language Learning, 53*(S1), 167–210. https://doi.org/10.1111/1467-9922.00227

Moskovsky, C., Assulaimani, T., Racheva, S., & Harkins, J. (2016). The L2 motivational self system and L2 achievement: A study of Saudi EFL learners. *The Modern Language Journal, 100*(3), 641–654. https://doi.org/10.1111/modl.12340

Noels, K. A., Clément, R., & Pelletier, L. G. (1999). Perceptions of teachers' communicative style and students' intrinsic and extrinsic motivation. *The Modern Language Journal, 83*(1), 23–34. https://doi.org/10.1111/0026-7902.00003

Ozdemir, E., & Papi, M. (2022). Mindsets as sources of L2 speaking anxiety and self-confidence: The case of international teaching assistants in the U.S. *Innovation in Language Learning and Teaching, 16*(3), 234–248. https://doi.org/10.1080/17501229.2021.1907750

Papi, M. (2010). The L2 motivational self system, L2 anxiety, and motivated behavior: A structural equation modeling approach. *System, 38*(3), 467–479. https://doi.org/10.1016/j.system.2010.06.011

Park, G. P. (1997). Language learning strategies and English proficiency in Korean university students. *Foreign Language Annals, 30*(2), 211–221. https://doi.org/10.1111/j.1944-9720.1997.tb02343.x

Peng, J. E., & Woodrow, L. (2010). Willingness to communicate in English: A model in the Chinese EFL classroom context. *Language Learning, 60*(4), 834–876. https://doi.org/10.1111/j.1467-9922.2010.00576.x

Pyun, D. O., Kim, J. S., Cho, H. Y., & Lee, J. H. (2014). Impact of affective variables on Korean as a foreign language learners' oral achievement. *System, 47*, 53–63. https://doi.org/10.1016/j.system.2014.09.017

Samimy, K. K., & Tabuse, M. (1992). Affective variables and a less commonly taught language: A study in beginning Japanese classes. *Language Learning, 42*(3), 377–398. https://doi.org/10.1111/j.1467-1770.1992.tb01341.x

Sato, M. (2020). Generating a roadmap for possible selves via a vision intervention: Alignment of L2 motivation and classroom behavior. *TESOL Quarterly, 55*, 427–457.

Sato, M., & Lara, P. (2019). Interaction vision intervention to increase second language motivation: A classroom study. In M. Sato, & S. Loewen (Eds.), *Evidence-based second language pedagogy: A collection of instructed second language acquisition studies* (pp. 287–313). Routledge.

StataCorp. (2019). *Stata statistical software: Release 16*. StataCorp LLC, College Station, TX.

Taheri, H., Sadighi, F., Bagheri, M. S., & Bavali, M. (2019). EFL learners' L2 achievement and its relationship with cognitive intelligence, emotional intelligence, learning styles, and language learning strategies. *Cogent Education, 6*(1), Article 1655882. https://doi.org/10.1080/2331186X.2019.1655882

Teimouri, Y., Goetze, J., & Plonsky, L. (2019). Second language anxiety and achievement: A meta-analysis. *Studies in Second Language Acquisition, 41*(2), 363–387. https://doi.org/10.1017/S0272263118000311

Toker, S., & Akbay, T. (2022). A comparison of recursive and nonrecursive models of attitude towards problem-based learning, disposition to critical thinking, and creative thinking in an computer literacy course for preservice teachers. *Education and Information Technologies, 27*, 6715–6751. https://doi.org/10.1007/s10639-022-10906-y

Wu, K. H. (2010). The relationship between language learners' anxiety and learning strategy in the CLT classrooms. *International Education Studies, 3*(1), 174–191.

Yashima, T. (2002). Willingness to communicate in a second language: The Japanese EFL context. *The Modern Language Journal, 86*(1), 54–66. https://doi.org/10.1111/1540-4781.00136

Zarrinabadi, N., Rezazadeh, M., Karimi, M., & Lou, N. M. (2022). Why do growth mindset make you feel better about learning and your selves? The mediating role of adaptibility. *Innovation in Language Learning and Teaching, 16*(3), 249–264. https://doi.org/10.1080/17501229.2021.1962888